PROFILE

Websites that work. Details that matter.

Mohamed Walaa Zain

Backend & Web Developer

I build web apps with clear interfaces, careful data rules, and results you can inspect.



THREE REASONS TO LOOK CLOSER

43%
Lower homepage load time

154
Backend tests in saved verification

205
Spreadsheet rows reconciled

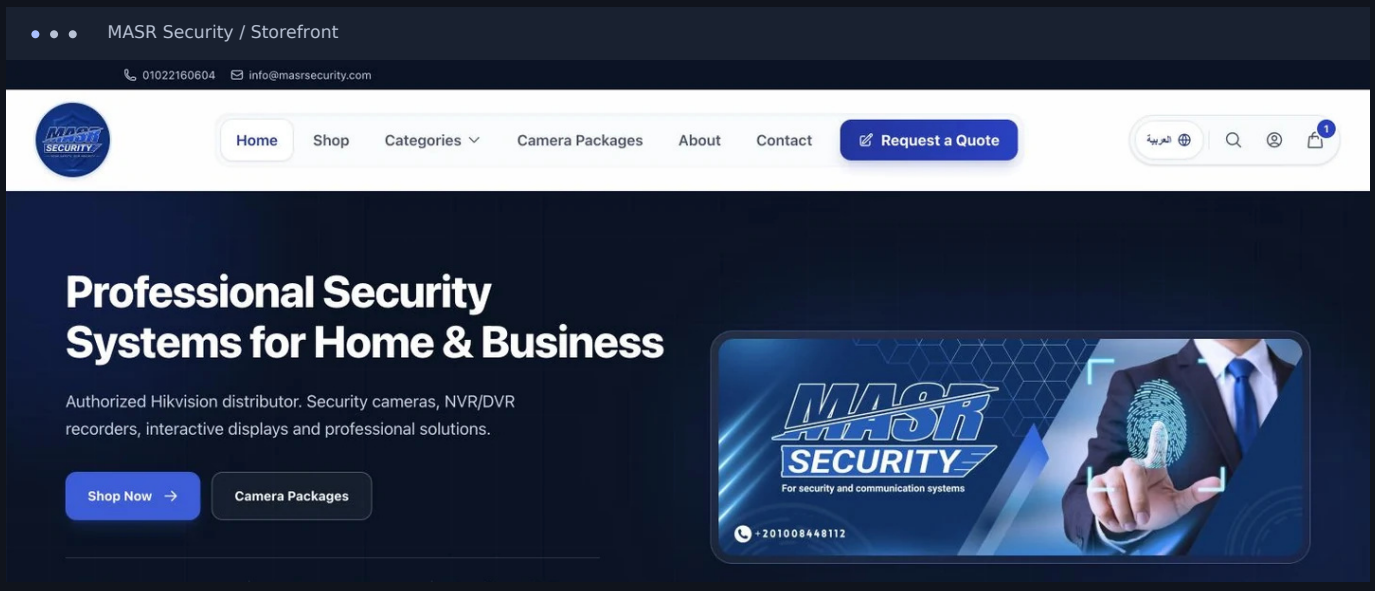
I turn requirements into working web applications. I check the rules and measure the changes.

mohamed.walaa.zain@gmail.com GitHub LinkedIn Portfolio website

MASR OVERVIEW

MASR Security

An Arabic and English website for security products. Customers can browse, build a cart, and check out. The manager can update the catalog and prices.



Real storefront screenshot. Arabic and English interface.

MY ROLE

I worked independently on development. I implemented requirements given by the manager.

Delivered client website	Arabic & English	Payments: sandbox verified

WHAT CHANGED

- Built the Arabic and English storefront, cart, checkout, and account pages.
- Added catalog tools, bulk pricing, previews, and undo rules.
- Calculated totals on the server. Stock and orders change together.
- Integrated Paymob and verified payment outcomes in sandbox only.
- Reduced image and page work, then measured loading again.

Visit storefront **View source on GitHub**

Paymob is integrated and verified in sandbox only. No payment-volume claim.

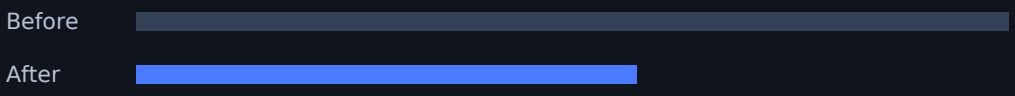
MASR IMPROVEMENTS

Faster. Lighter. Measured.

I reduced image size and page work. Then I measured the same pages again. All 15 MASR JavaScript logic tests passed again on 20 October 2026.

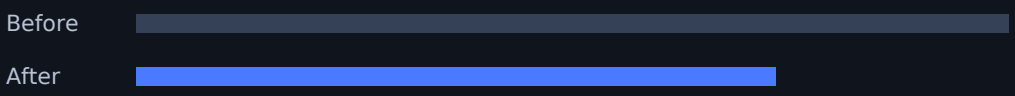
Faster homepage loading

43%



Faster shop loading

27%



5 Oct 2026 mobile lab medians. Home: 3 before and 3 after runs. Shop: 2 successful before and 3 after runs. One before run failed. These results do not describe every visitor.

Mobile Lighthouse scores: home 92 to 99; shop 98 to 100. 5 October 2026 lab medians.

52%

Smaller page styling

5 Oct 2026. Uncompressed files: 119,910 to 57,940 bytes. About 52% smaller.

787

Better-sized images

5 Oct 2026. 191 originals kept. 787 WebP versions generated. These are file counts.

11 → 8

Less database work

Undated earlier test. Home 11 to 8; warm dashboard 5 to 0. Five isolated samples.

205

A cleaner product catalog

4 Oct 2026. 205 rows; 107 products removed; 6 referenced parts kept. Units differ.

Earlier query tests used isolated sample data. They do not measure current production requests.

WHY I BUILT IT THIS WAY

Send images that fit the screen

Keep originals. Generate WebP sizes. Separate shop and dashboard styling.

Tradeoff More files need storage. Each visitor can load a better-sized image.

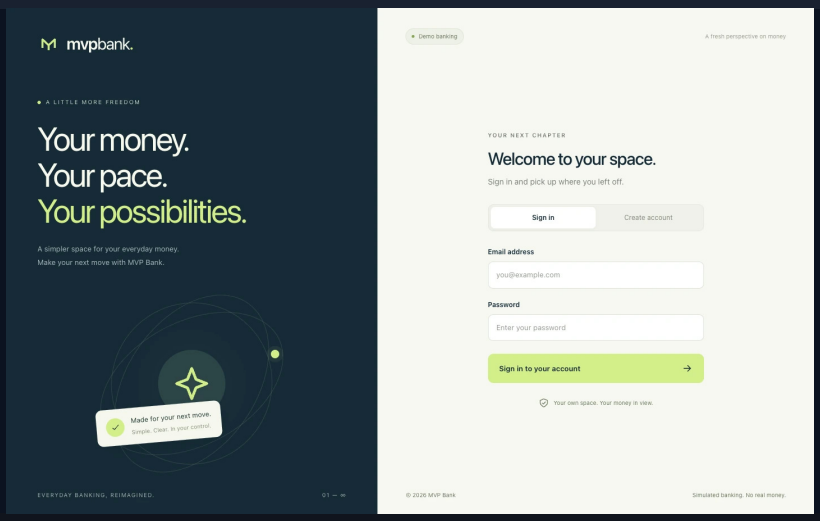
MVP BANK

One request. One money change.

A simulated bank. It handles exact amounts, checks access, and keeps repeated or overlapping requests from changing money twice.

I built the project, its money rules, access rules, and request recovery. I wrote checks for the API and browser logic.

MVP Bank / Sign-in screen



Actual MVP Bank sign-in screen. Demo access stays protected.

Simulated banking | Exact amounts | Safe retries

154	615	7	17
backend tests	individual checks	overlap / locking cases	documented app actions

Saved backend verification: 17 Oct 2026. Locking audit: 28 Sep 2026. Backend suite not rerun. All 49 frontend tests passed again on 20 Oct 2026.

WHY I BUILT IT THIS WAY

Store exact amounts

Store amounts as whole piastres, so EGP 1 is stored as 100.

Tradeoff Input rules are stricter. Money avoids decimal rounding errors.

Make retries safe

Keep the first receipt with its request key. A retry gets the same result.

Tradeoff The browser must keep the key. A retry cannot debit the account again.

Hosted demo · sign-in protected

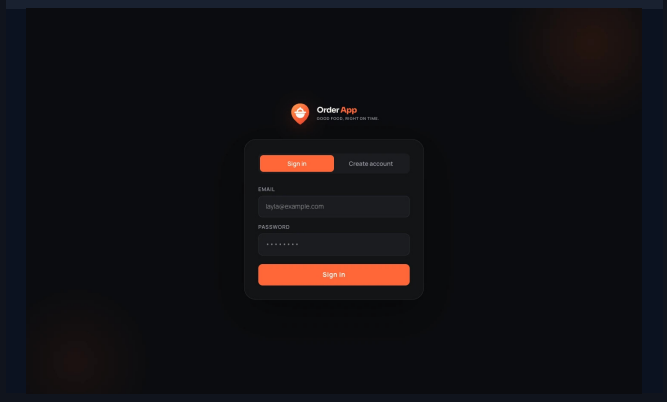
This is a simulated bank. It does not hold real deposits. No public source link is available.

ORDER APP & LEARNING ARCHIVE

The right action, by the right person.

A food-ordering project with clear rules for customers, restaurant owners, and admins. Each action checks the role, the record owner, and the order state.

Order App / Sign-in screen



Actual Order App frontend sign-in screen. No customer records shown.

MY ROLE

I built the API, access rules, owner applications, order flow, and scoped delivery tracking.

- 3 roles
- 5 order states
- Ownership checks

WHAT CHANGED

Role	Actions
Customer	Place and view own orders. Cancel while pending.
Restaurant owner	Manage owned restaurants and meals. Update their orders.
Admin	Review owner applications and oversee the app.

THE ORDER FLOW

Pending → **Preparing** → **On the way** → **Delivered**

Pending | **Cancelled**

View source on GitHub

LEARNING ARCHIVE

BATMAN CAVE IDEAS

An early Laravel project for saving and sharing ideas. I used it to learn sign-in, validation, notifications, and basic data actions.

Edit and delete ownership checks still need protection. Learning archive only. No live demo or security claim.

View source on GitHub

SKILLS, EDUCATION & CONTACT

How I build and learn.



Backend & Web Developer

I build web applications with PHP, Laravel, SQL, and JavaScript. I turn requirements into clear screens and careful data rules. I worked independently on MASR Security, following requirements from its manager. My own projects help me test harder problems.

Jobs and freelance work are welcome. Remote, hybrid, and on-site work are open for discussion.

mohamed.walaa.zain@gmail.com

GitHub **LinkedIn**

TOOLS I USE

Build

PHP / Laravel / SQL / Vanilla JavaScript / HTML & CSS / Blade

Connect

REST APIs / Laravel Sanctum / API Resources / OpenAPI / Paymob integration / Signed webhooks

Protect data

MySQL / PostgreSQL / Eloquent ORM / Database transactions / Row locking / Authorization policies / Validation

Verify & improve

PHPUnit / Pest / Frontend logic tests / Git & GitHub / Postman / Performance measurement / OOP & SOLID fundamentals

EDUCATION

B.Sc. in Artificial Intelligence

Kafrelsheikh University | 2023

COURSEWORK · SELF-DIRECTED STUDY

PHP For Beginners
Laravel From Scratch (2026 Edition)
SOLID Principles in PHP
Leveraging AI for Laravel Development
Object-Oriented Principles in PHP
Laravel API Master Class
Design Patterns in PHP